

PROBLEMS IN DENSITY MATRICES

On the slides we showed the partial traces of the pure Bell state $|\Psi^+\rangle_{AB} = (|00\rangle + |11\rangle)/\sqrt{2}$ resulted in maximally mixed $\rho_A = \rho_B = I_2/2$.

① Now take a mixed system AB where,

$$\rho_{AB} = \frac{1}{2} |00\rangle\langle 00| + \frac{1}{2} |01\rangle\langle 01|$$

$\Rightarrow$ we are mixed to start with.

Problem: Find ρ_A & ρ_B and compare.

$$i) \rho_{AB} = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rho_A = \text{tr}_B \rho_{AB}$$

$$= \begin{bmatrix} \text{tr} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} & \text{tr } 0 \\ \text{tr } 0 & \text{tr } 0 \end{bmatrix}$$

$$\rho_A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\rho_B = \text{tr}_A \rho_{AB}$$

$$= \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\rho_B = \frac{1}{2} I_2$$

$$ii) \text{tr} \rho_A^2 = \text{tr} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = 1$$

① $\Rightarrow$ a pure state
- not mixed.

$$\text{e.g. } |\Psi\rangle_A = |0\rangle$$

$$\text{tr} \rho_B^2 = \text{tr} \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} = \frac{1}{2}$$

$$\text{e.g. } \{|0\rangle, \rho_0 = \frac{1}{2}\}, \{|1\rangle, \rho_1 = \frac{1}{2}\}.$$

② $\Rightarrow$ a mixed state.

$$iii) \text{ Makes sense } \because \rho_{AB} = \frac{1}{2} |00\rangle\langle 00| + \frac{1}{2} |01\rangle\langle 01|$$

$\Rightarrow$ A is always in $|0\rangle$ state. B is 50:50 $|0\rangle, |1\rangle$.